

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Periodic Classification of Elements

Formative-only topic in CBSE 2026-27: useful for school assessment; not year-end board assessed

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

State the modern periodic law.

Answer:

The properties of elements are periodic functions of their atomic numbers. Arranging elements by increasing atomic number places elements with similar properties in recurring groups.

Question 2

Why do elements in one group often have similar chemical properties?

Answer:

They have the same number of valence electrons in the outermost shell. This gives them similar tendencies to gain, lose, or share electrons.

Question 3

How does metallic character generally change across a period and down a group?

Answer:

It generally decreases from left to right across a period and increases down a group. Across a period atoms tend to hold electrons more strongly; down a group the outer electrons are farther from the nucleus.

Question 4

What is the central idea or skill tested in Periodic Classification of Elements?

Answer:

The central focus is: The properties of elements are periodic functions of their atomic numbers. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Periodic Classification of Elements.

Answer:

First point: The properties of elements are periodic functions of their atomic numbers. Second point: They have the same number of valence electrons in the outermost shell. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Periodic Classification of Elements lead to an important result or conclusion?

Answer:

Begin with the key idea: The properties of elements are periodic functions of their atomic numbers. Then connect it to the result: They have the same number of valence electrons in the outermost shell.

Periodic Classification of Elements - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Periodic Classification of Elements?

Answer:

Use this supporting detail: The properties of elements are periodic functions of their atomic numbers. Arranging elements by increasing atomic number places elements with similar properties in recurring groups. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Periodic Classification of Elements.

Answer:

They have the same number of valence electrons in the outermost shell. This gives them similar tendencies to gain, lose, or share electrons. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Periodic Classification of Elements?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: It generally decreases from left to right across a period and increases down a group.

Question 10

Compare the first and second important ideas in Periodic Classification of Elements.

Answer:

The first idea is: The properties of elements are periodic functions of their atomic numbers. The second idea is: They have the same number of valence electrons in the outermost shell. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Periodic Classification of Elements.

Answer:

Cause or starting point: The properties of elements are periodic functions of their atomic numbers. Effect or outcome: They have the same number of valence electrons in the outermost shell. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Periodic Classification of Elements?

Answer:

A clear sequence is: The properties of elements are periodic functions of their atomic numbers. Next, They have the same number of valence electrons in the outermost shell. Finally, It generally decreases from left to right across a period and increases down a group.

Periodic Classification of Elements - Continued

Question 13

Give a 40-50 word answer outline for an important question from Periodic Classification of Elements.

Answer:

Start with the direct answer: The properties of elements are periodic functions of their atomic numbers. Add one supporting point: They have the same number of valence electrons in the outermost shell. End with the outcome or message: It generally decreases from left to right across a period and increases down a group.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Periodic Classification of Elements?

Answer:

Use an accurate scientific statement: The properties of elements are periodic functions of their atomic numbers. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Periodic Classification of Elements?

Answer:

Connect the idea to its reason or result. For instance: They have the same number of valence electrons in the outermost shell. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Periodic Classification of Elements.

Answer:

A sound inference is that it generally decreases from left to right across a period and increases down a group. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Periodic Classification of Elements.

Answer:

Apply the concept by first stating: The properties of elements are periodic functions of their atomic numbers. Then justify the response with: They have the same number of valence electrons in the outermost shell.

Question 18

What would make an answer from Periodic Classification of Elements incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: The properties of elements are periodic functions of their atomic numbers. Also add: It generally decreases from left to right across a period and increases down a group.

Periodic Classification of Elements - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Periodic Classification of Elements.

Answer:

Assertion: The properties of elements are periodic functions of their atomic numbers. Reason: They have the same number of valence electrons in the outermost shell. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Periodic Classification of Elements in two clear steps?

Answer:

Step 1: state the concept or event - The properties of elements are periodic functions of their atomic numbers. Step 2: support it with the explanation or result - They have the same number of valence electrons in the outermost shell.

Question 21

What answer structure is suitable for a 3-mark question from Periodic Classification of Elements?

Answer:

Use three connected points: The properties of elements are periodic functions of their atomic numbers. They have the same number of valence electrons in the outermost shell. It generally decreases from left to right across a period and increases down a group. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Periodic Classification of Elements?

Answer:

Give an introduction, develop these key points - The properties of elements are periodic functions of their atomic numbers. They have the same number of valence electrons in the outermost shell. - and close with: It generally decreases from left to right across a period and increases down a group. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Periodic Classification of Elements be applied to a new situation or case study?

Answer:

Identify the relevant principle first: The properties of elements are periodic functions of their atomic numbers. Then use the chapter's reasoning to explain the case: They have the same number of valence electrons in the outermost shell.

Periodic Classification of Elements - Continued

Question 24

Which detail from Periodic Classification of Elements is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: It generally decreases from left to right across a period and increases down a group. Explain it in context instead of using it as an isolated statement.

Question 25

Write a brief concluding statement for an answer on Periodic Classification of Elements.

Answer:

A suitable conclusion is: It generally decreases from left to right across a period and increases down a group. This conclusion should follow naturally from the explanation rather than introduce a new point.

Question 26

Make a three-point revision note for Periodic Classification of Elements.

Answer:

1. The properties of elements are periodic functions of their atomic numbers. 2. They have the same number of valence electrons in the outermost shell. 3. It generally decreases from left to right across a period and increases down a group. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Periodic Classification of Elements before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - The properties of elements are periodic functions of their atomic numbers. - and supported it with a reason, example, calculation, or outcome such as: They have the same number of valence electrons in the outermost shell.

Question 28

Why does Periodic Classification of Elements remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: The properties of elements are periodic functions of their atomic numbers. They have the same number of valence electrons in the outermost shell. It generally decreases from left to right across a period and increases down a group.

Question 29

Create a final 60-80 word answer plan for a long question from Periodic Classification of Elements.

Answer:

Opening: The properties of elements are periodic functions of their atomic numbers. Development: They have the same number of valence electrons in the outermost shell. Conclusion: It generally decreases from left to right across a period and increases down a group. Use complete sentences and include all required evidence, steps, or references.

Periodic Classification of Elements - Continued

Question 30

What is one last-minute revision question you should ask yourself about Periodic Classification of Elements?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: The properties of elements are periodic functions of their atomic numbers. and They have the same number of valence electrons in the outermost shell.

Sources: [CBSE Science curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

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